T3000 API & Protocol

Control Messages Index · built Nov 13, 2026

Control Messages Index

Control messages are how the web interface communicates with the T3000 platform. Each message performs a specific operation like reading device data, updating values, or managing graphics.

Quick Reference: Jump to any message documentation below.

All Control Messages

Message	Type	Purpose	Common Use
GET_PANEL_DATA	Data Retrieval	Load all cached data for a panel	Initial panel view load
GET_INITIAL_DATA	Graphics	Load graphics screen data	Opening graphics editor
LOAD_GRAPHIC_ENTRY	Graphics	Load specific graphic entry	Graphics screen display
SAVE_GRAPHIC_DATA	Graphics	Save graphics screen changes	Saving graphics edits
GET_WEBVIEW_LIST	Data Retrieval	Read specific entry type with range	Loading inputs, outputs, programs, etc.
UPDATE_WEBVIEW_LIST	Data Update	Write values to device entries	Updating setpoints, outputs, variables
UPDATE_ENTRY	Data Update	Update single entry field	Quick field update
GET_PANELS_LIST	Discovery	Get all online panels	Device list refresh
GET_ENTRIES	Data Retrieval	Get multiple specific entries	Batch data retrieval

Messages by Category

Data Retrieval Messages

Purpose: Reading device data and configuration

- **GET_PANEL_DATA** - Load all cached panel data
- **GET_WEBVIEW_LIST** - Read specific entry types (inputs, outputs, programs, etc.)
- **GET_ENTRIES** - Read multiple specific entries by index

Typical Flow:

1. Frontend requests data via Rust API
2. C++ checks cache first
3. If cache miss, reads from device via BACnet
4. Returns JSON response

Data Update Messages

Purpose: Writing values and configurations to devices

- **UPDATE_WEBVIEW_LIST** - Update multiple entries in bulk
- **UPDATE_ENTRY** - Update single entry field

Typical Flow:

1. Frontend sends new values
2. C++ validates data
3. Writes to device via BACnet
4. Updates local cache
5. Returns success/failure

Graphics Messages

Purpose: Managing graphics screens

- **GET_INITIAL_DATA** - Initial graphics screen load
- **LOAD_GRAPHIC_ENTRY** - Load specific graphics entry
- **SAVE_GRAPHIC_DATA** - Save graphics changes

Graphics Storage:

- Stored as ZIP files in `DatabaseImages/` folder
- Contains PNG images and metadata
- Linked to device panel and entry index

Discovery Messages

Purpose: Finding devices and panels

- **GET_PANELS_LIST** - List all online panels

- **GET_ENTRIES** - Query available entries

Message Format

Request Structure

JSON

```
{
  "action": 17,  // Message type number
  "source": 1,   // Source: 0=T3000, 1=WebUI
  "panelId": 0,
  "serialNumber": 12345,
  // ... message-specific parameters ...
}
```

Response Structure

JSON

```
{
  "action": "MESSAGE_TYPE_RES",  // Response action name
  "data": {
    // ... response data ...
  }
}
```

Error Response

JSON

```
{
  "action": "ERROR_RES",
  "error": {
    "code": -1,
    "message": "Error description",
    "details": "Additional context"
  }
}
```

Implementation Location

File: `T3000-Source/T3000/BacnetWebView.cpp`

Main Handler Function:

```cpp
void CBacnetWebView::HandleWebViewMsg(const CString& receivedMessage)
{
    // Parse JSON
    Json::Value json;
    // ...

    int action = json.get("action", Json::nullValue).asInt();

    switch (action)
    {
        case WEBVIEW_MESSAGE_TYPE::GET_PANEL_DATA:
            // Handle message...
            break;
        case WEBVIEW_MESSAGE_TYPE::GET_WEBVIEW_LIST:
            // Handle message...
            break;
        // ... more cases ...
    }
}
```

Entry Types Reference

Used with GET_WEBVIEW_LIST and UPDATE_WEBVIEW_LIST:

Type	Value	Name	Max Count	Description
BAC_IN	1	Inputs	64	Analog/digital inputs
BAC_OUT	2	Outputs	64	Analog/digital outputs
BAC_VAR	3	Variables	64	User variables
BAC_PRG	6	Programs	16	Control programs
BAC_TL	10	Trendlogs/Monitors	12	Data logging
BAC_PID	11	Controllers	16	PID control loops
BAC_SCH	12	Schedules	32	Weekly schedules
BAC_HOL	13	Holidays	32	Annual holidays
BAC_GRP	23	Graphics	50	Graphics screens
BAC_ALM	9	Alarms	64	Alarm points

Common Parameters

panelId

- Type: Integer
- Range: 0-254
- Description: Device panel number in network

serialNumber

- Type: Integer (unsigned)
- Range: 0-4294967295
- Description: Device unique serial number

objectinstance

- Type: Integer
- Range: 0-4194303
- Description: BACnet object instance ID

entryType

- Type: Integer
- Range: 1-23

- Description: Type of entry (see Entry Types table)

entryIndexStart / entryIndexEnd

- Type: Integer
- Range: 0 to (maxCount - 1)
- Description: Range of entries to read/write

Response Times

Typical response times:

- **Cached data:** < 50ms
- **Fresh read (1-10 items):** 200-500ms
- **Fresh read (full range):** 1-5 seconds
- **Bulk update:** 500ms-3 seconds

Error Codes

Code	Meaning	Common Cause
-1	Read timeout	Device offline or not responding
-2	Invalid parameter	Out of range index or invalid type
-3	Cache miss	No cached data available
-4	Write failed	Device rejected write command
-5	Object mismatch	Wrong device responded

See Also

- Platform Overview →
- BACnet Commands →
- Data Structures →